Positional effects in Japanese imperfect puns

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1. An Introduction

An overarching issue:

Are linguistic patterns purely formal and computational or are they affected by other cognitive factors to a significant degree?

We show:

- In forming imperfect puns, Japanese speakers care about psycholinguistic prominence of different positions in the words.
- Speakers avoid disparities between corresponding segments in psycholinguistically prominent positions.
- We find a non-trivial parallel in phonological patterns—the preservation of phonological contrasts in prominent positions.

2. Background and hypothesis

Dajare: imperfect puns in Japanese

Japanese speakers create sentences using two similar sounding words or phrases:

(1) **buta**-ga **buta**reta 'A pig was hit'.
(2) **futon**-ga **futton**da 'A futon was blown away'.

Two corresponding elements can be non-identical:

(3) Ai**z**u-san-no ai**s**u 'Ice from Aizu'
(4) Okosa**m**a-o okosa**n**aide 'Don't wake up a kid'

Kawahara & Shinohara (2009) show that two corresponding elements must be **psychoacoustically similar**.

If speakers attempt to maximize the similarity between the corresponding elements, then would it matter where a mismatch is located? It would because:

A mismatch appearing in word-initial or word-final positions would be more perceptually—or cognitively—salient than a mismatch appearing in word-medial positions, because materials at the word edges are more important for speech processing.

(See Horowitz, White, & Atwood 1968; Horowitz, Chilian & Dunigan 1969; Gupta, 2005; Gupta, Lipinski, Abbs & Lin, 2005 among many others).

3. Corpus Study

We first started with a corpus study. The corpus is based on Tajiya, 2001 and Kawahara & Shinohara, 2009 (itself based on data from three websites and native speaker consultation), which contained 145 examples of three-syllable long puns with one mismatched consonant pair. In this corpus, we counted the numbers of matched and mismatched syllables in initial, medial, and final positions.

	Initial	Medial	Final	Total
Mismatch	64	43	38	145

Observations:

1.Medial syllables are more likely to contain mismatches than final syllables.
2.But medial syllables are LESS likely to contain mismatches than initial syllables.

➔ **Maybe a more controlled experiment is desirable.**

4. Experiment

Method

Stimuli: Sentences consisting of pairs of words that minimally differ in voicing ([k-g], [g-k], [s-z], [z-s], [t-d] , [d-t]).

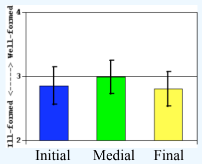
The mismatch was placed in either initial, medial or final syllable. For example:

(i) Initial mismatch: **s**asetsu-ni **z**asetsu 'I gave up turning left.'
(ii) Medial mismatch: hi**s**ashi-ni hi**z**ashi 'Sunlight at a roof.'
(iii) Final mismatch: furii**s**u-ga furii-**z**u 'Sweater got frozen.'

Task: Judged the wellformedness of the stimuli as *dajare*. But we first asked how funny they are, so that the participants would separate the wellformedness of sentences as *dajare* from how funny they are.

Participants: 37 speakers of Japanese participated in this study. We excluded 8 speakers from the results because they did not judge typical examples of *dajare* as *dajare* in the practice session.

Result



Within-subject contrast analyses:

Initial vs. Medial: $t(28)=2.5$, $p=.012$
Medial vs. Final: $t(28)=3.48$, $p<.001$
Initial vs. Final: $t(28)=1.16$, $p=.23$

Discussion

1.Speakers dislike mismatches at word-initial and final positions.
2.We do not observe a difference between word-initial and final positions (cf. Horowitz et al. 1968, 1969; Nootbaum 1981 for evidence that word-initial segments are psycholinguistically more prominent than word-final segments).

5. Parallel with phonological mapping

A recent view of the mapping from Underlying Representation (UR) to Surface Representation (SR) in Correspondence Theory (McCarthy and Prince 1995).

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/ a  b  c  d  e /   Underlying Representation
  :  :  :  :  :                                   UR-SR mapping
[ a     c  f  e ]   Surface Representation
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(i) Input segments and output segments stand in correspondence. (ii) A pair of corresponding segments is required to be as similar as possible by faithfulness constraints.

We can straightforwardly extend this notion of Correspondence to pun pairing.

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[ a  b  c  d  e ]   Word 1       Pun pairing
  :  :  :  :  :
[ a     c  f  e ]   Word 2       (a kind of surface-to-surface correspondence)
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It so happens that in phonological mapping, speakers disfavor disparities in word-initial positions (Beckman 1998) as well as in word-final positions (Walter 2002), just as we found in our experiment. Correspondence is in general subject to positional effects.

6. CONCLUSIONS

In judging the wellformedness of *dajare*, Japanese speakers disfavor mismatches in positions that are psycholinguistically prominent (word edges).

We find a non-trivial parallel between our findings and phonological patterns.

We thus suspect that these external cognitive factors affect—or even shape—our linguistic behaviors to a non-negligible extent (see also Kawahara, Shinohara, & Uchimoto 2008 for initial syllable effects on sound symbolism).

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